CAVS OFF-ROAD PROVING GROUND · AUTONOMOUS VEHICLE TESTING

Post-Run Review Interface

Dashboard sketches · v1

Use case	Reviewing a completed autonomy run from a recorded ROS 2 bag
Target display	12″ tablet, landscape, browser UI driven by Raspberry Pi 5
Data in these sketches	Values taken from the provided bag autonomy_demo_0 (72.1 s, 124,845 messages)
Pages in this set	1 Dashboard overview · 2 Event review · 3 Fault drill-down · 4 Navigation

Zeenat Islam

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1 · Dashboard — run overview

Landing view after a run is loaded. Playhead parked at the end of the run.

CAVS · Run Review `autonomy_demo_0 ▾` 11 Sep 2026, 11:45 CDT · NATURE / MTX stack · 3 cams · Ouster LiDAR — Stopped by operator

RUN SUMMARY

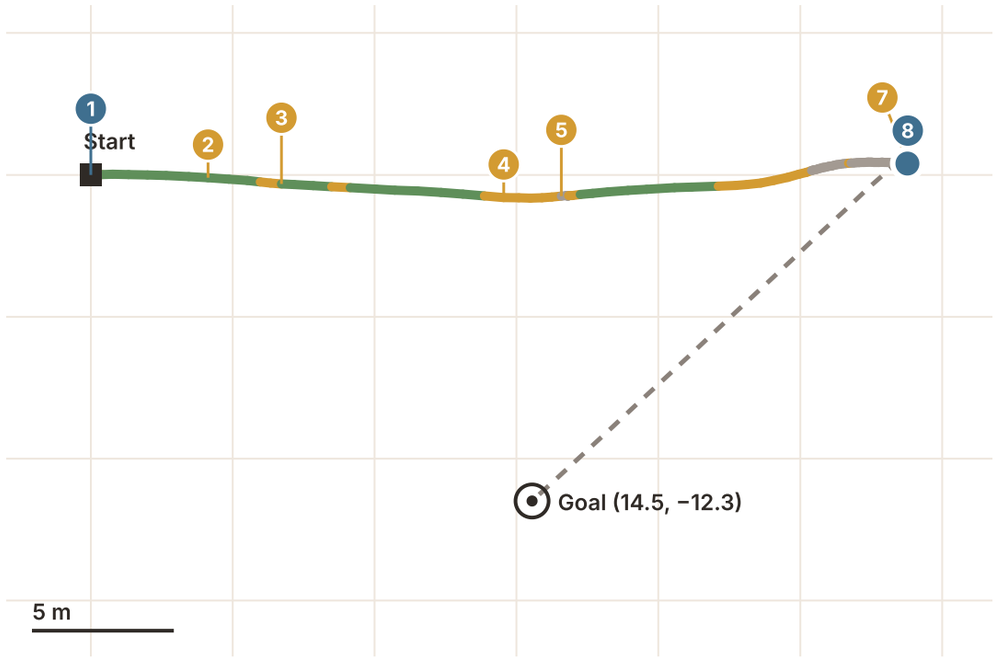
Duration **72.1 s** 11:45:48 → 11:47:00	Distance **29.2 m** driven
Autonomy **56.5 s** 8.2 s → 64.7 s	Max speed **2.2 m/s** at 15.8 s
Interventions **1** operator STOP	Goal **Not reached** 2 waypoints sent

KEY EVENTS

1	8.1s	Autonomy engaged
2	15.5s	Controller patience exceeded
3	17.9s	Cam 2 frame gap 265 ms
4	23.2s	Slowdown zone
5	25.7s	Controller patience exceeded
6	64.7s	STOP pressed, e-stop latched
7	65.8s	Cam 2 frame gap 170 ms
8	67.5s	E-stop cleared

ROUTE OVERVIEW

Open in Path Planning ›

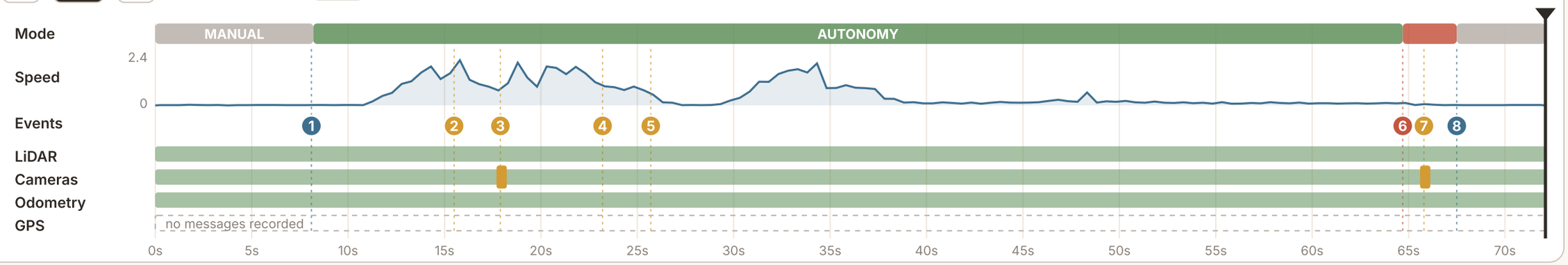


SYSTEM HEALTH · WHOLE RUN

LiDAR `/ouster/points`	10.0 Hz	no dropouts
Camera 0 `/cam_0/image_raw`	19.1 Hz	no gaps
Camera 1 `/cam_1/image_raw`	19.1 Hz	no gaps
Camera 2 `/cam_2/image_raw`	18.9 Hz	2 frame gaps
Odometry `/dlio/odom`	99.9 Hz	no gaps
IMU `/ouster/imu`	99.8 Hz	no gaps
GPS `no topic in log`		no messages

2 camera gaps · 0 LiDAR dropouts · – GPS Faults ›

⏮ ▶ ⏭ 72.1 / 72.1 s 1× ▾ Tap an event or drag the playhead



Dashboard | Path Planning | Faults | Sensors | Run Logs

A Top bar names the run and states the outcome in one chip, so the reviewer knows how the run ended before reading anything else.

B Run summary answers “how far, how long, was the goal reached, did a person step in.”

C Route trace is coloured by speed; the dashed leg is the plan that was never driven after the STOP.

D Numbered event pins match across the event list, the map and the timeline.

E System health shows each sensor for the whole run, colour coded green / amber / red / grey.

F Timeline lanes stack mode, speed, events and sensor streams on one shared time axis.

2 · Dashboard — reviewing an event

Reviewer taps pin 2 on the timeline. Every panel jumps to that moment.

CAVS · Run Review `autonomy_demo_0` ▾ 11 Sep 2026, 11:45 CDT · NATURE / MTX stack · 3 cams · Ouster LiDAR — Stopped by operator

EVENT 2 OF 8 — ‹ prev · next ›

2 Controller patience exceeded

t = 15.5 s · controller_server · WARN/ERROR

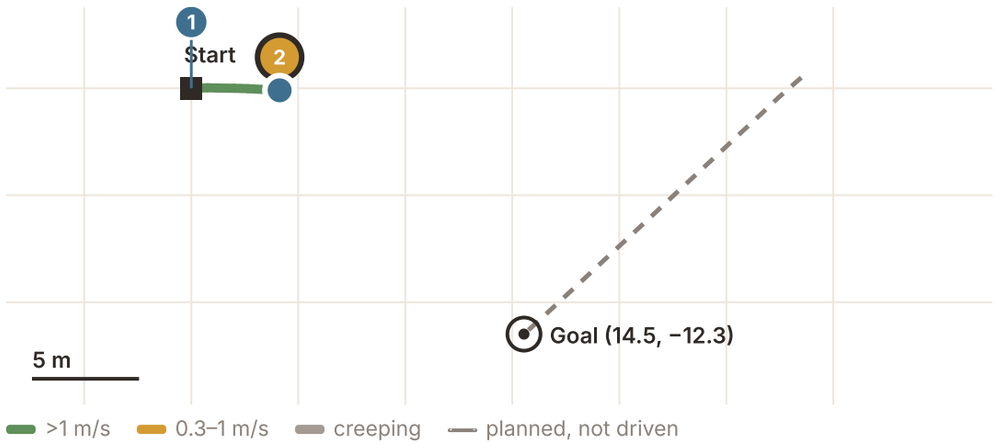
Mode	AUTONOMY
Speed	1.59 m/s
Position	(4.1, -0.1) m
Preceded by	11× “collision ahead” in 0.5 s
Outcome	follow_path aborted
Sensors	all nominal

/rosout around this moment

```
15.0 WARN collision ahead! (×11)
15.5 ERROR patience exceeded
15.5 WARN follow_path aborted
15.6 WARN loop 8.3 Hz (target 20)
```

Open in Run Logs ›

ROUTE UP TO 15.5 S — Open in Path Planning ›



CAMERA FRAMES AT PLAYHEAD — Sensors ›

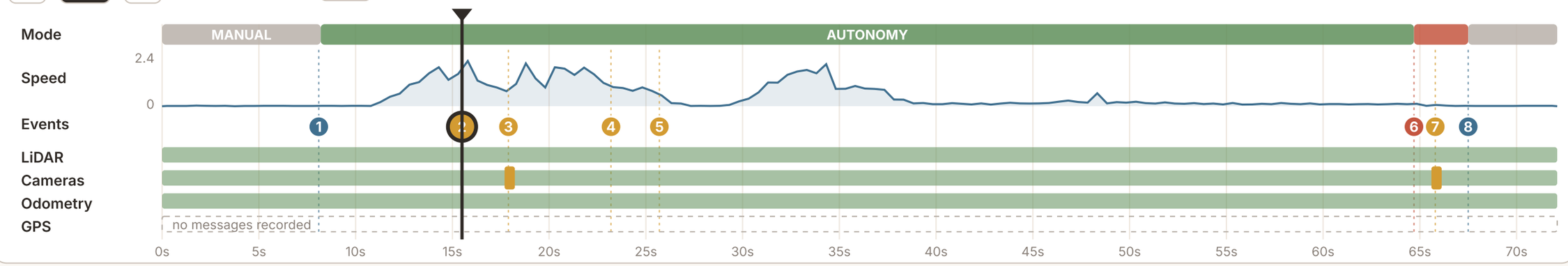


SYSTEM HEALTH · WHOLE RUN

LiDAR	/ouster/points	10.0 Hz, no dropouts
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Camera 1	/cam_1/image_raw	19.1 Hz, no gaps
Camera 2	/cam_2/image_raw	18.9 Hz, 2 frame gaps
Odometry	/dlio/odom	99.9 Hz, no gaps
IMU	/ouster/imu	99.8 Hz, no gaps
GPS	no topic in log	no messages

2 camera gaps · 0 LiDAR dropouts · – GPS — Faults ›

⏮ ▶ ⏭ 15.5 / 72.1 s 1× ▾ — Tap an event or drag the playhead



Dashboard | Path Planning | Faults | Sensors | Run Logs

A Left column swaps the summary for an event card: what happened, vehicle state at that instant, and the log lines around it.

B Map trims the route to the playhead so the reviewer sees where the vehicle was, not where it ended up.

C Camera frames from all three cameras at the same timestamp sit under the map.

D Prev / next steps through events in order, so a full review needs no scrubbing.

3 · Dashboard — fault drill-down

Reviewer taps the amber Camera 2 tile. The dashboard opens a short fault summary in place.

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RUN SUMMARY

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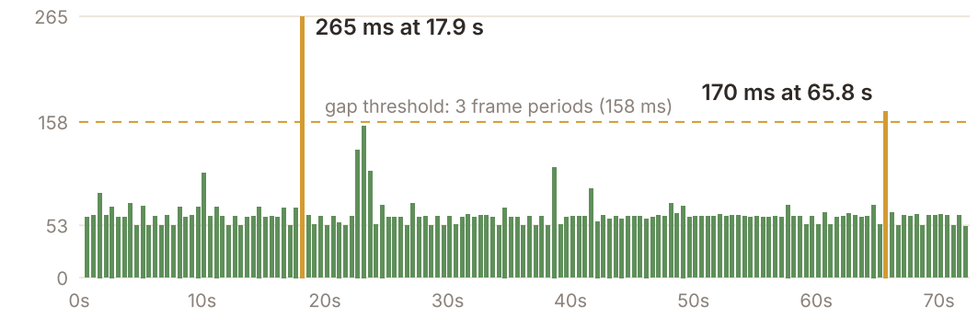
KEY EVENTS

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CAMERA 2 · FRAME GAPS

Open in Faults ›

Longest frame interval per 0.5 s (ms)



2	265 ms	18.9 Hz	19.0 Hz
gaps	longest	mean rate	expected

OTHER FAULT CLASSES IN THIS RUN

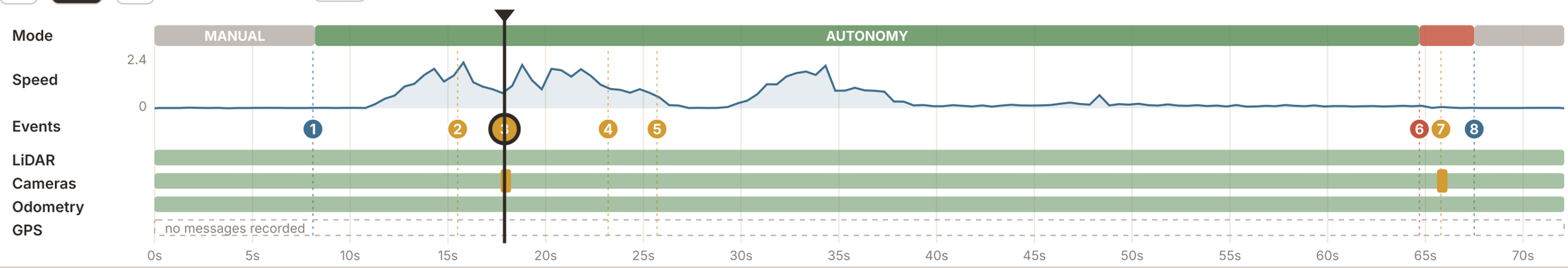
LiDAR dropout	none
GPS dropout	no GPS stream in log
Camera 0 / 1 lag	none

SYSTEM HEALTH · WHOLE RUN

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Camera 2 `/cam_2/image_raw`	18.9 Hz	2 frame gaps
Odometry `/dlio/odom`	99.9 Hz	no gaps
IMU `/ouster/imu`	99.8 Hz	no gaps
GPS `no topic in log`		no messages

2 camera gaps · 0 LiDAR dropouts · – GPS — Faults ›

⏮ ▶ ⏭ **17.9 / 72.1 s** 1× ▾ — Tap an event or drag the playhead



Dashboard | Path Planning | Faults | Sensors | Run Logs

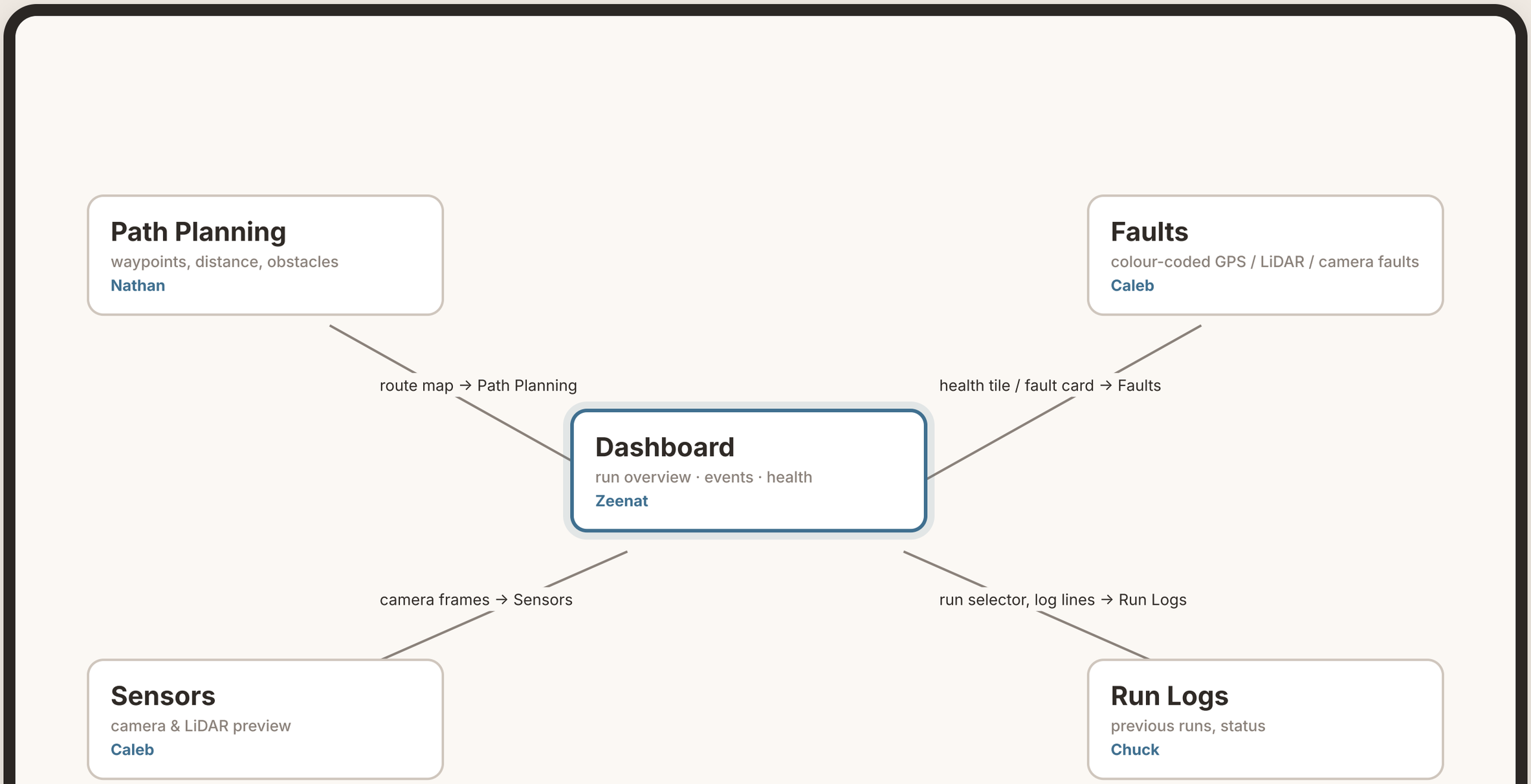
A Tapping any amber or red health tile opens its fault summary in the centre panel and moves the playhead to the first occurrence.

B The same three fault classes the course bag injects (GPS dropout, LiDAR dropout, camera lag) always have a slot, even when a run has none.

C Detailed fault analysis lives on the Faults page; the dashboard only answers “did it happen, when, how bad.”

4 · Navigation

Where the dashboard hands off to the other pages in the team's design.



Shared across every page: run selector and outcome chip (top), timeline playhead time, bottom menu bar

A The dashboard is the hub. Each panel links to the page that owns its detail.

B The selected run and playhead time carry over when moving between pages, so the reviewer never loses their place.